

Track 1 – Vector Implementation of Snell's Law

➤ Objective

The goal of this project is to implement the vector form of Snell's Law in order to compute the refracted ray direction when light passes between two different media. This implementation is suitable for ray tracing and rendering engines.

➤ Mathematical Model

When a light ray passes from medium n_1 to medium n_2 , Snell's Law states:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

For rendering engines, we use the **vector form** of Snell's Law:

$$T = \eta I + (\eta \cos \theta_i - \sqrt{1 - \eta^2 (1 - \cos^2 \theta_i)}) N$$

Where:

I = Incident unit vector

N = Surface normal unit vector

T = Refracted vector

$$\eta = n_1 / n_2$$

$$\cos \theta_i = -I \cdot N$$

➤ Vectorization Using Dot Product

Instead of using trigonometric functions directly, we compute:

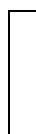
$$\cos \theta_i = -I \cdot N$$

➤ Test Case Validation

We tested the model using:

$$I = (0, -1, 0)$$

$$N = (0, 1, 0)$$



$n_1 = 1.0$ (air)

$n_2 = 1.5$ (glass)

$$\eta = 1/1.5 = 0.66$$

The computed refracted vector shows bending toward the normal, which is physically correct when entering a denser medium.

➤ Implementation

The algorithm follows these steps:

1. Normalize vectors I and N
2. Compute $\eta = n_1/n_2$
3. Compute $\cos\theta_i = -I \cdot N$
4. Compute: $k = 1 - \eta^2(1 - \cos^2\theta_i)$
5. If $k < 0$, Total Internal Reflection occurs.
6. Otherwise compute the refracted vector T

The algorithm has constant complexity $O(1)$, since it performs a fixed number of vector operations.

➤ Visualization

The simulation renders a 3D glass cube and a laser beam interacting with its surface.

Red ray: Incident ray (incoming laser beam)

Green ray: Refracted ray inside the cube

Yellow ray: Exit ray after leaving the cube

Vertical green line: Surface normal vector

Transparent blue cube: Glass medium

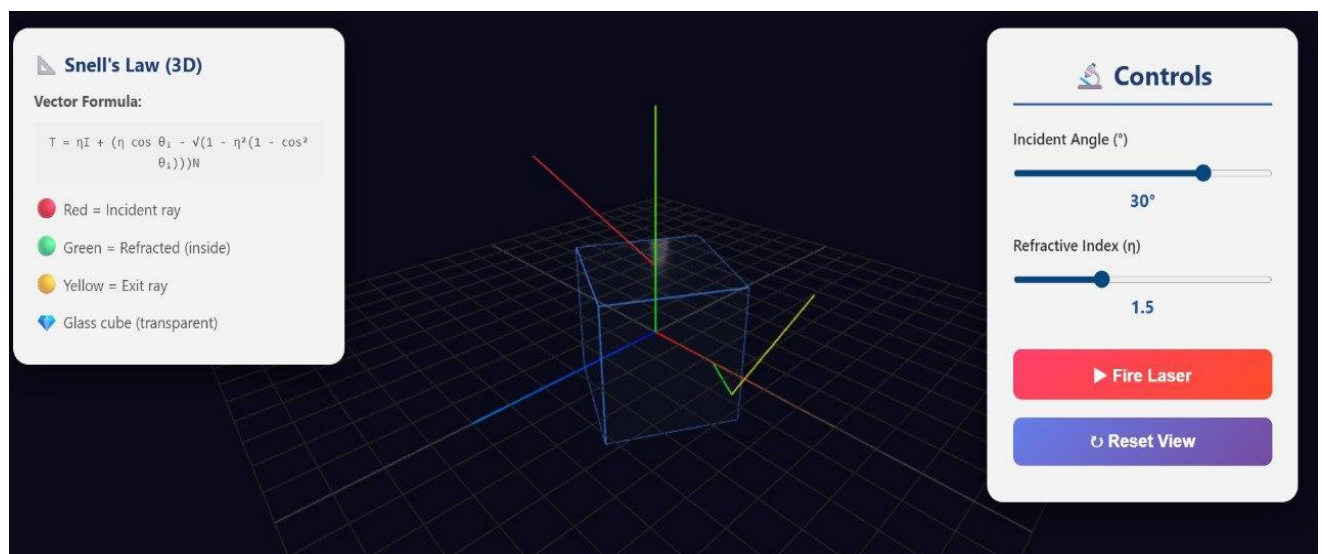
*Incident angle ($^\circ$)

*Refractive index (η)

When the refractive index is increased (e.g., $\eta = 1.5$), the refracted ray bends toward the normal, which is physically correct when light enters a denser medium (air $\rightarrow$ glass).

The 3D axes (X, Y, Z) and grid are displayed to provide spatial reference.

The visualization confirms the correctness of the vector implementation of Snell's Law.



➤ **Discussion**

The simulation follows Snell's Law correctly.

When light enters the glass cube ($n = 1.5$), the ray bends toward the normal, and bends away when exiting back to air.

The ray is refracted twice: at the air–glass interface and at the glass–air interface.

Small numerical errors may appear due to floating-point precision.

The vector formulation is efficient and suitable for real-time rendering.

➤ **Snell's Law Visualization Schema**

